

Morphology-Conditioned Physics-Based Motion Imitation Across 128 Body Shapes

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Abstract

[TODO: Write last, after Section 5 results are final.] We present a single physics-based control policy that imitates arbitrary human motions across 128 distinct SMPL body shapes (26–144 kg, 1.13–1.67 m), in contrast to prior physics-based motion trackers that are restricted to a single fixed morphology. Our approach combines a two-stage curriculum (neutral pretraining followed by shape transfer) that decouples motion out-of-distribution generalization from shape out-of-distribution generalization, trained at full HumanML3D scale (20,946 clips). We introduce a shape-invariant, DOF-angle-space evaluation protocol that isolates skill consistency from body-scale confounds, and show that a single policy achieves consistent joint-articulation tracking across the full shape distribution.

1 Introduction

Physics-based motion tracking policies (e.g., PHC, PULSE, ExBody2, GMT) reproduce reference human motions under physical simulation with high fidelity, but are trained and evaluated on a single, fixed body morphology. This limits their applicability to settings that require one controller to generalize across a population of body shapes — e.g., animating diverse virtual characters or transferring a single policy across a fleet of robots with varying morphology.

We target a single policy that imitates arbitrary human motions across a wide distribution of body shapes: 128 SMPL shapes (64 betas $\times$ 2 genders), spanning 26–144 kg and 1.13–1.67 m. The central difficulty is that a naive combination of *motion diversity* (tens of thousands of clips) and *shape diversity* (128 morphologies) confounds two independent axes of generalization: whether the policy has seen a given *motion*, and whether it has seen a given *body*. Conflating them in one training/eval split makes failures hard to diagnose and progress hard to attribute.

Contribution. We propose a two-stage curriculum — neutral pretraining on the full 20,946-clip HumanML3D library at a single body shape, followed by shape-transfer training across 128 shapes — that explicitly decouples motion-OOD from shape-OOD. We identify and fix two structural failure modes specific to the shape-transfer stage (residual PD control, morphology observation normalizer reset; Section 3), and we introduce an evaluation protocol (Section 5) that measures shape-invariant skill consistency directly, rather than relying on world-space error or single-body baselines that structurally cannot support a cross-shape generalization claim.

2 Related Work

Physics-based motion tracking. PHC, PULSE, ExBody2, GMT, and other H2O-style trackers learn high-fidelity imitation policies under physical simulation, but all assume a single, fixed-

morphology character; none address generalization across a body-shape distribution.

Morphology-conditioned RL. HUMOS [TODO: cite] conditions motion generation (via shape-conditioned diffusion) on SMPL body shape, producing per-shape-adapted reference motions rather than a naively retargeted, shape-invariant one. MorFiC [TODO: cite] [TODO: one-line summary]. Our work builds directly on HUMOS’s motion data to supply a shape-*dependent* training signal (Section 3), but targets a control policy rather than a motion generator.

Mixture-of-experts in actor-critic RL. [TODO: cite prior MoE-in-RL work]. We evaluate MoE as an architecture candidate for handling motion diversity (Section 4) and find it does not separate from a capacity-matched dense baseline in our setting.

3 Method

3.1 Data & Asset Pipeline

Our pipeline spans three components: ProtoMotions (RL training framework), a HUMOS fork (shape-conditioned motion diffusion [TODO: cite]) for shape-adapted motion targets, and SMPLSim for per-shape body assets. We sample 128 body shapes as 64 SMPL shape vectors $\beta \sim \mathcal{U}[-3, 3]^{10}$ (seed 46) $\times$ 2 genders, each keyed by a deterministic hash. For each shape, SMPLSim generates one MJCF asset (physically accurate inertial mass, simplified feet, frozen hands, mass-scaled PD gains). *Frame-0 grounding* computes each shape’s lowest collision point once per shape (not per motion, $\sim 64\times$ cheaper) and shifts every motion’s root height so it starts just above the ground plane.

Why HUMOS. Naively retargeting AMASS motion capture onto a new body shape (plain forward kinematics, same joint-angle trajectory reposed onto a different skeleton) produces a reference motion that is *exactly shape-invariant* by construction — the DOF-angle target is identical regardless of body shape, and therefore carries **zero shape-adaptation gradient**. HUMOS instead resamples each clip via shape-conditioned diffusion, producing a target that genuinely differs by body shape. Data pipeline: AMASS (18 of 23 sub-datasets) $\rightarrow$ SMPL+H forward kinematics at 20 fps $\rightarrow$ mocap-artifact cleaning (treadmill/skating removal) $\rightarrow$ HumanML3D 263-dim feature extraction $\rightarrow$ HUMOS diffusion inference per (clip, shape) pair. A known caveat, central to the reward design in Section 3.5 and Section 4.5: HUMOS output is measurably noisier/jitterier than ground-truth AMASS mocap.

3.2 Problem Formulation

The policy $\pi_\theta(a_t \mid s_t)$ outputs joint target actions given proprioceptive state, future reference poses, previous action, and a morphology-conditioning vector morphology_obs (gender id + raw SMPL betas, 11-dim; see Section 3.7), all under a shared running normalizer (~ 1014 -dim total observation). A scalar *phase* variable $\phi = t/T_{\text{clip}} \in [0, 1]$ (elapsed motion time over clip length) is appended to resolve temporal aliasing in periodic motions — e.g., a squat and a kneel are visually near-identical at matching points in their own cycle without phase information. Contact-reward bodies are feet + knees + wrists (extended from feet-only) so that contact_match_rew gives nonzero signal on crawl/kneel clips.

3.3 Two-Stage Curriculum

Stage 1 (neutral pretrain): full 20,946-clip library, single neutral body shape ($\beta = 0$). **Stage 2 (shape transfer):** same motion library, 128 body shapes. The split is motivated by an earlier finding that a single combined train/eval split conflates motion-OOD and shape-OOD; decoupling into two stages lets each isolate one generalization axis.

Two structural fixes were required for Stage 2 to work at all:

1. **Residual PD control.** Stage 1’s PD target is a fixed neutral pose offset by the policy action, $q_{\text{target}} = q_{\text{neutral}} + s \cdot \tanh(a)$. Naively reusing this in Stage 2 produced 3–4 $\times$ higher action jerk. We instead anchor the PD target to the *reference* pose at the current timestep:

$$q_{\text{target}}(t) = q_{\text{ref}}(t) + 0.3 \cdot \tanh(a_t). \quad (1)$$

2. **Morphology-observation normalizer reset.** Stage 1’s constant- β training drives the running variance of the morphology-observation slice toward zero; the resulting normalizer then clamps Stage 2’s real per-shape betas to a near-binary ± 5 signal. We reset the mean/variance statistics for just that observation slice before Stage 2 training.

Stage 1 alone reaches 84.9% success at epoch 20,200, with an 8.7% persistent-failure tail dominated, at this full 20,946-clip scale, by a *single-leg dynamic-balance* failure category not visible at smaller (1,024-clip) scale.

3.4 Architecture

The final architecture is a morphology-conditioned wide-MLP trunk: actor 2896×6 , critic 1024×4 . This choice was reached by iterative comparison against two alternatives (mixture-of-experts, self-attention temporal encoder), both null results reported in Section 4.3; we state only the final architecture here.

3.5 Reward and Termination Design

We use the “discover” reward relaxation: drop tracking-error-threshold termination (keep only fall detection) and drop effort/smoothness/contact-timing reward terms, keeping only tracking-shaping rewards. The motivation (CMU getting-up-policy-style two-phase curriculum) is to relax every constraint except “did you get anywhere near the reference,” so the policy’s only objective on the hardest clips is finding *any* successful trajectory. This followed four independent earlier attempts (mass-scaled PD gains, per-segment mass-scaled gains, soft-tracking relaxation, AMP-style discriminator reward) that converged to the same 40–65% success band with no lever separating from the others.

World-space tracking rewards. For body position/rotation/velocity/angular velocity and root height, each reward is an exponential of mean squared (or angular) error:

$$r_{gt} = \exp(c_{gt} \cdot \text{mean} \|p - p_{\text{ref}}\|^2), \quad c_{gt} = -25.0 \text{ to } -100.0 \quad (2)$$

$$r_{gr} = \exp(c_{gr} \cdot \text{mean} \angle(q, q_{\text{ref}})^2), \quad c_{gr} = -5.0 \quad (3)$$

$$r_{gv} = \exp(c_{gv} \cdot \text{mean} \|v - v_{\text{ref}}\|^2), \quad c_{gv} = -0.5 \quad (4)$$

$$r_{gav} = \exp(c_{gav} \cdot \text{mean} \|\omega - \omega_{\text{ref}}\|^2), \quad c_{gav} = -0.1 \quad (5)$$

$$r_{rh} = \exp(c_{rh} \cdot (h - h_{\text{ref}})^2), \quad c_{rh} = -100.0 \quad (6)$$

where p, q, v, ω are world-space rigid-body position, rotation, linear velocity, and angular velocity, and $\angle(\cdot, \cdot)$ is quaternion angular difference.

DOF-space (shape-invariant) tracking rewards. Local joint-angle position/velocity rewards, the shape-invariant counterpart to r_{gt}/r_{gv} (Section 5):

$$r_{dp} = \exp(c_{dp} \cdot \text{mean} \|q_{\text{dof}} - q_{\text{dof,ref}}\|^2), \quad c_{dp} = -40.0 \quad (7)$$

$$r_{dv} = \exp(c_{dv} \cdot \text{mean} \|\dot{q}_{\text{dof}} - \dot{q}_{\text{dof,ref}}\|^2), \quad c_{dv} = -0.5 \quad (8)$$

and a yaw-only heading reward, $r_{\text{heading}} = \exp(c_{\text{heading}} \cdot \Delta\psi^2)$ with $c_{\text{heading}} = -10.0$, where $\Delta\psi$ is the (wrapped) root-heading difference — near-shape-invariant world-frame grounding, since a pelvis turn means about the same thing regardless of body proportions.

The reference reward bundles used in this work:

- **World-space bundle:** weights $(w_{gt}, w_{gr}, w_{gv}, w_{gav}, w_{rh}) = (0.5, 0.3, 0.1, 0.2, 0.2)$, coefficients $(-25.0, -5.0, -0.5, -0.1, -100.0)$.
- **DOF-space bundle:** weights $(w_{dp}, w_{dv}, w_{\text{contact}}, w_{\text{heading}}, w_{rh}) = (0.8, 0.3, -0.1, 0.1, 0.2)$, coefficients $(-40.0, -0.5, -, -10.0, -100.0)$.

The DOF-space bundle reached 75–78% success, the largest single gain in this lineage, with near-zero fall-terminations. Combining world- and DOF-space signal across two motion sources (AMASS-canonical, shape-invariant by construction; HUMOS, shape-dependent) is the subject of the in-progress reward-source design in Section 4.5.

3.6 Training Infrastructure at Scale

Full-scale training streams per-clip motion files from object storage (R2) rather than loading one static file (`GlobalClipPool`): a per-rank deterministic resident training pool with UCB-style priority refresh, a fixed evaluation holdout immune to resident-set churn, a weight floor (solved clips retain nonzero rehearsal priority), and a random rehearsal fraction.

3.7 Morphology Conditioning Representation

The final morphology-conditioning representation is raw SMPL betas concatenated with gender, $\text{morphology_obs} = [\text{gender_id}, \beta_1, \dots, \beta_{10}]$ (11-dim). This was chosen after comparing against three alternatives (Section 4.4): FiLM-style multiplicative conditioning, a learned shape-embedding, and a 15-dim z -scored physics-derived feature vector (mass, height, limb-length-style scalars).

4 Experiments

4.1 Setup

Dataset scale evolved through the project: a 1,024-clip $\times$ 128-shape pilot $\rightarrow$ Stage 1’s full 20,946-clip library at one neutral shape $\rightarrow$ a 20,946-clip $\times$ 2-shape intermediate set (population-median male/female bodies, used to stress-test MoE without waiting on full 128-shape data) $\rightarrow$ a 150-clip $\times$ 128-shape reduced-scale testbed adopted for cost reasons as the standard architecture/reward ablation corpus (Sections 4.3–4.5) $\rightarrow$ full 20,946-clip $\times$ 128-shape training via `GlobalClipPool` (Section 3.6) for the final run.

Metrics: `dp_error` (DOF-angle, shape-invariant — primary metric, full definition and justification in Section 5), `gt_error/ gr_error` (world-space position/rotation, scale-confounded across shapes, kept for legacy/reward-design tracking), success rate, held-out clip evaluation.

4.2 Main Results

Stage 1 neutral pretrain reaches 84.9% success at epoch 20,200 (~ 174 h, 6 GPUs), with an 8.7% persistent-failure tail (Section 3.3). For Stage 2 shape-transfer, a residual-adaptor lineage (frozen backbone + adapter, v1–v6) plateaued at 78–82% success across every variant tried and was abandoned in favor of a from-scratch Stage 2 trunk, still in progress [TODO: final Stage 2 success numbers pending]. Stage 2 results should be read alongside the shape-consistency metrics of Section 5, not success rate in isolation.

4.3 Architecture Evolution: Flat MLP $\rightarrow$ MoE $\rightarrow$ Self-Attention

Presented as a narrative of iterative design — the actual chronological order tried, each step motivated by the failure mode of the last — validated at reduced scale (150-clip / 1,024-clip corpora) rather than full scale, for cost reasons.

1. **Flat-MLP baseline.** Wide MLP over flat-concatenated observations (history/lookahead frames concatenated, not tokenized). Establishes a ~ 75 –80% success-rate ceiling on the hardest-clip subset.
2. **Mixture-of-Experts (MoE).** Hypothesis: distinct motion classes (crawl/kneel/squat/turning) benefit from routed specialist sub-policies. **Result: null** — a capacity-matched (parameter-matched, $\sim 95\%$) wide-MLP control caught up to the MoE-stable variant on all metrics; whether routing helps remains an open question.
3. **Self-attention temporal encoder.** Hypothesis: tokenizing per-frame history/ lookahead and attending (reusing the existing **Transformer** module) helps structure learning over the dilated frame sequence. **Result: null** — no separation from flat-concat at matched training steps.

Both alternatives were null/inconclusive at the scale tested; we retain the simpler flat-MLP architecture, carried forward as a bundled recipe into the one full-scale run rather than re-validated at full scale.

4.4 Morphology Conditioning Ablation: Raw Betas vs. Physics-Derived Features

Pilot-scale comparison ($1,024\text{-clip} \times 128\text{-shape}$ corpus): raw SMPL betas (21,400 epochs) vs. a 15-dim z -scored physics-derived feature vector (mass, height, limb-length-style scalars; 17,200 epochs). Persistent-failure-clip counts (177 vs. 167) are **not statistically significant**; qualitative visual review favored physics features (5/8 vs. 0/8 clips judged visually improved). Raw betas are kept as the simpler default for the full-scale run — the physics-feature signal was not strong enough to justify the added feature-engineering surface at $20,946 \times 128$ scale.

4.5 AMASS/HUMOS Reward Source Design

[TODO: In progress, not yet resolved.]

Motivation: AMASS-canonical DOF-space targets are exactly shape-invariant (Section 3.1), so training on them alone gives zero shape-conditioning gradient; HUMOS is the only channel whose target genuinely differs by shape. Iterative design, same framing as Sections 4.3–4.4:

1. DOF-space-only bundle on the canonical corpus — clean `dp_error` convergence, world-space metrics flat.
2. Episode-level source-switching (hard-mask reward terms by AMASS-vs-HUMOS per episode) — world-space metrics improve over (1) but underperform the unmasked world-space baseline at matched steps (63% vs. 83% success), likely because masking halves how often the world-space reward fires per rollout.
3. Combined, unmasked, full-factor reward on the mixed corpus with per-source *weights* (not term masks) — outcome pending.

4.6 Reward/Termination Ablation

Discover-relaxed vs. a stricter baseline configuration (full termination + effort/smoothness/ contact-timing terms), validated at reduced scale for the same cost reason as Sections 4.3–4.4, then carried forward as-is to full scale by explicit decision.

4.7 Failure Analysis

Persistent-failure motion clusters show substantial overlap across single-shape and multi-shape settings: 97.1% overlap between the full-scale-scratch hardest-clip scoreboard and the single-shape neutral persistent-failure set — the plateau is an intrinsic motion-difficulty ceiling, not shape- or reward-specific. Shape-failure correlation (Pearson $r \approx -0.09$ to 0.003 between shape extremity and failure) is reported here in full; the headline framing of the same result appears in Section 5.

4.8 Cross-Shape Generalization

Held-out beta interpolation/extrapolation (experiment E7) is the strongest form of Section 5’s shape-consistency claim, extending it past the 128 training shapes to unseen interpolated/extrapolated shapes. [TODO: Blocked on an `infer.py` held-out-beta pipeline bug; deferred in favor of reusing already-existing held-out-shape data once a Stage 2 checkpoint is ready.]

5 Evaluation

Framing. Success rate against a single-body external baseline (PHC/PULSE/ExBody2/GMT) cannot demonstrate this paper’s actual claim — that one policy learns the *same motion* across a wide body-shape distribution — since those methods only ever evaluate one body. World-space error (`gt_error`, meters) is not a fair axis even for *internal* cross-shape comparison: a taller body produces larger absolute position deviations for the same relative motor error, so raw meters bake in a body-scale confound before measuring skill at all. DOF-angle-space error (`dp_error`, radians) is exactly shape-invariant by construction (same joint target regardless of body) and is therefore the

primary lens for the paper’s core claim, not a secondary metric introduced only for the reward-design work of Section 4.5. Formally,

$$\text{dp_error} = \text{mean}_d |q_{\text{dof}}[d] - q_{\text{dof,ref}}[d]|, \quad \text{gt_error} = \text{mean}_b \|p[b] - p_{\text{ref}}[b]\|_2, \quad (9)$$

where d ranges over actuated DOFs (radians) and b ranges over rigid bodies (meters).

Scope note: `dp_error` **measures joint rotation, not root trajectory.** q_{dof} is the actuated joint-angle vector, local to each body’s own kinematic tree — hinge/spherical joint values relative to their parent link. It excludes the root free-joint (pelvis translation and orientation in the world), which is tracked separately via `gt_error`/`gr_error` and is exactly the shape-dependent signal deliberately excluded from the headline metric. The shape-invariance claim built on `dp_error` is therefore scoped to **limb articulation** — “the same joint rotates the same way regardless of body” — not to whole-body world-space behavior. A tall and a short body executing the same clip show identical `dp_error` trajectories even though their root paths differ in absolute distance (shorter legs, shorter stride, same joint angles); that is expected and outside what this metric measures. A root-level shape-adaptation claim (e.g., stride length scaling correctly with leg length) would require a separate, explicitly shape-normalized root metric (e.g., root displacement normalized by height/leg length) — raw `gt_error` cannot support that claim either, for the same body-scale-confound reason.

5.1 Cross-Shape Consistency per Clip

For each clip, we report the spread (std/IQR) of `dp_error` across the 128 training shapes. A tight spread demonstrates shape-independent skill execution; a wide spread would undercut the claim.

5.2 Shape-Failure Correlation

Computed across multiple runs (full detail in Section 4.7): Pearson $r \approx -0.09$ to 0.003 between shape extremity (mass/height) and failure — effectively zero relationship. This is the headline evidence that body shape does not predict success, i.e. the same motion is being learned regardless of body.

5.3 Held-Out Shape Generalization

Extends the consistency claim past the 128 training shapes to interpolated/extrapolated unseen shapes (experiment E7, Section 4.8) — the strongest version of this evaluation. [TODO: Currently blocked on the `infer.py` pipeline bug.]

5.4 External Comparison (Footnote)

Neutral-shape success rate against single-body baselines (PHC-style, $\beta = 0$) is retained only as a minor sanity-check — confirming neutral-shape quality was not sacrificed for shape generalization — not as a headline result, since a single-body number structurally cannot speak to the shape-generalization claim this section exists to make.

6 Discussion / Limitations

A capability ceiling on a cluster of hard motions (crawl/kneel/squat, single-leg dynamic balance) persists across shape, reward-relaxation, architecture, and morphology-representation interventions.

What did not work and why: FiLM-style conditioning (failed outright, traced to a 12,288-output conditioner bottleneck and multiplicative-gradient instability); residual adapters (v1–v6, plateaued 78–82%); MoE routing (no separation from a capacity-matched control); self-attention temporal encoding (no separation from flat-concat; the shared **Transformer** module also lacks positional encoding — flagged, deliberately not fixed, judged unlikely to move the ceiling); physics-derived morphology features (no statistically significant gain over raw betas, though qualitatively suggestive).

The broader lesson from the design-evolution narrative (Sections 4.3–4.4) is that the hard-motion ceiling looks like it is about *motion difficulty*, not about which architecture or which conditioning representation carries the signal — every lever tried moved metrics by less than noise. Resource constraints limited ablation breadth to reduced-scale corpora, addressed directly in the framing of those sections rather than hidden.

The shape-invariance evaluation (Section 5) is a separate axis from this ceiling discussion: the ceiling is about which *motions* are hard, not about *which bodies* execute them — consistent with the near-zero shape-failure correlation (Section 4.7).

7 Conclusion

[TODO: Write last, after Sections 4.2, 4.5, and 5 are final.]